Summary in Graph

Exam Summary (GO Classes CS Test Series | C-Programming | Topic Wise Test 1)

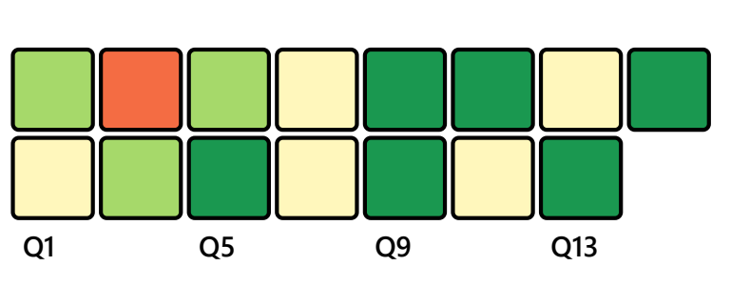
58.7%
14.67/25

9 Correct | 3 Wrong | 3 Skipped | 45m of 45m

Qs. Attempted:	12 5 + 7	**Correct Marks:**	15 3 + 12	**Total Questions:**	15 5 + 10
Correct Attempts:	9 3 + 6	**Penalty Marks:**	0.33 0.33 + 0	**Total Marks:**	25 5 + 20
Incorrect Attempts:	3 2 + 1	**Resultant Marks:**	14.67 2.67 + 12	**Exam Duration:**	45 Minutes
				Time Taken:	45 Minutes

Individual Question Statistics

View Order: Sorted Order ▾ Select Metric: Marks Obtained ▾



* Average Stats based on first-attempt stats of Top 10% (178) out of 1775 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C | Medium (35%)

Consider the k bit binary pattern. $\mathrm{T_{max}}$ and $\mathrm{T_{min}}$ are maximum and minimum signed numbers we can represent using k bits. $\mathrm{U_{max}}$ and $\mathrm{U_{min}}$ are maximum and minimum unsigned numbers we can represent using k bits. Numbers are represented using 2's complement system.

Which of the following(s) is/are true for $k = 16$?

Here $|.|$ represents absolute value of a number i.e. $|r| = -r$ if $r < 0$ otherwise $|r| = r$.

A. $|\mathrm{T_{min}}| = \mathrm{T_{max}} + 1$
B. $\mathrm{U_{max}} = 2 * \mathrm{T_{max}} + 1$
C. $\mathrm{U_{max}} = |\mathrm{T_{min}}| + \mathrm{T_{max}} + 1$
D. $\mathrm{U_{min}} = |\mathrm{T_{min}}|$

Your Answer: A;B | Correct Answer: A;B | Correct | Time taken: 08min 04sec | Discuss

Q #2 Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C | Hard (15%)

Consider Figure A and Figure B, which represent 4 bit signed and unsigned numbers, respectively, in the $2's$ complement system.

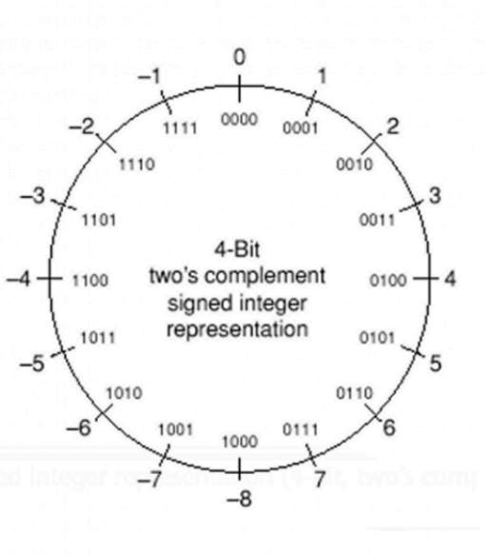


Figure A

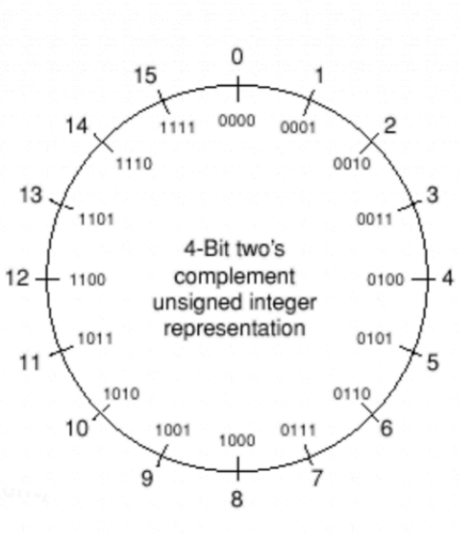


Figure B

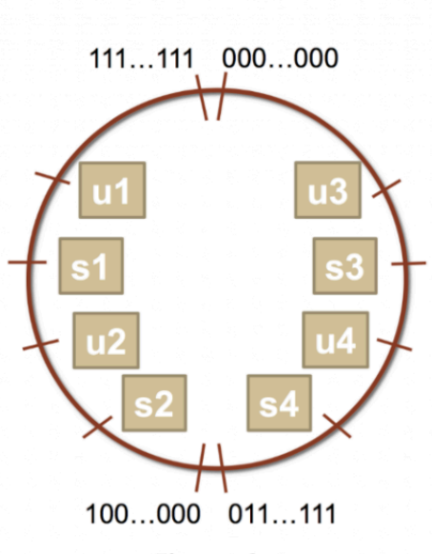


Figure C

Assume that a few variables are defined below and initialized in such a way that places them in the spot shown in Figure C.

```
int s1, s2, s3, s4;
unsigned int u1, u2, u3, u4;
```

Which of the following(s) is/are FALSE?

A. $s2 > s4$
B. $s1 > s2$
C. $s1 > u3$
D. $s3 > u3$

Your Answer: A;B;C | Correct Answer: A | Incorrect | Time taken: 02min 11sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C | Medium (56%)

Assume that x is a two's complement binary integer whose absolute value is not large (so no overflow problems). At right are four C expressions that use the $<<$ (shift left) and $\sim$ (bitwise complement: flip each 0 to 1 and each 1 to 0 in the binary representation) operators. For example, $7 << 1$ is 14 and ~ 0 is -1.

Match each expression to one of the mathematical functions at left.

$a.$	$4x$	I.	$(\sim x) + 1$
$b.$	$-x$	II.	$(x << 1) + x$
$c.$	$3x$	III.	$\sim x$
$d.$	$-x-1$	IV.	$x << 2$

A.
  a - IV
  b - I
  c - II
  d - III
B.
  a - IV
  b - III
  c - II
  d - I
C.
  a - II
  b - I
  c - IV
  d - III
D. None of these

Your Answer: C | Correct Answer: A | Incorrect | Time taken: 04min 39sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C | Easy (75%)

What will be the output of the following program?

```
#include<stdio.h>
main()
{
    char grade='A';
    int result=0;
    switch(grade)
    {
        case 'A':
            result+=4;
        case 'B':
            result+=3;
        case 'C':
            result+=2;
        default:
            result+=1;
    }
    printf("%d", result);
}
```

A. 10
B. 4
C. 9
D. 0

Your Answer: A | Correct Answer: A | Correct | Time taken: 00min 45sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C | Medium (45%)

What is the logical condition under which the following while loop will terminate?

```
int Beta = 5;
while (Beta > 0 && Beta < 10)
{
    printf("%d", Beta);
    scanf("%d", &Beta);
}
```

A. Beta < 0 && Beta $>= 10$
B. Beta < 0 || Beta > 10
C. Beta $<= 0$ || Beta $>= 10$
D. Beta < 0 || Beta $>= 10$

Your Answer: C | Correct Answer: C | Correct | Time taken: 01min 28sec | Discuss

Q #6 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Medium (34%)

What will be output on the execution of the following code segment?

```
#include<stdio.h>
main()
{
    unsigned num1 = 1;
    signed num2 = -1;
    if (num1 < num2)
        printf("less");
    else if (num1 > num2)
        printf("greater");
    else if (num1 == num2)
        printf("equal");
}
```

A. greater
B. less
C. equal
D. Error

Your Answer: B | Correct Answer: B | Correct | Time taken: 02min 46sec | Discuss

Q #7 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C | Hard (9%)

Imagine that our system uses a 5-bit integer representation and does addition and subtraction using the rules for 5-bit, two's complement arithmetic.

Which of the following is/are TRUE?

When converting to decimal, you should treat the value as either signed or unsigned according to the rules of C, where $\mathrm{T_{min}}$ and $\mathrm{T_{max}}$ are signed, as are plain constants, but a constant ending in U is unsigned.

A. -$\mathrm{T_{max}} - 1\mathrm{U}$ is 16 in decimal
B. -$\mathrm{T_{max}} - \mathrm{T_{min}}$ is 00001 in binary
C. $\mathrm{T_{max}} + \mathrm{T_{max}}$ is -2 in decimal
D. $\mathrm{T_{min}} - 1\mathrm{U}$ is 15 in decimal

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 44sec | Discuss

Q #8 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Hard (11%)

What will be the output of the given program?

```
#include<stdio.h>
void main()
{
    int sum =1;
    for ( unsigned int i=3; i>=0; --i)
    {
        if(i==0) sum = sum+1;
        else sum = sum *i;
    }
    printf("%d", sum);
}
```

A. 1
B. 7
C. 0
D. None of these

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 04min 09sec | Discuss

Q #9 Multiple Select Type | Award: 2 | Penalty: 0 | Programming | Hard (15%)

Which of the following condition(s) is/are evaluated to true for a given declaration of integer i.

```
int i = -7;
```

A. $-10 < i < -1$
B. $i \;\&\; 0 \mid 0 \;\&\; 1$
C. $0 \;\&\; 0 \;\&\; 0 \mid i$
D. $!(1 > 0 > i \;\&\; !i)$

Your Answer: C;D | Correct Answer: C;D | Correct | Time taken: 05min 13sec | Discuss

Q #10 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C | Medium (30%)

The statement

```
while ( --counter >= 1 )
    counter % 2 ? printf("A") : printf("B");
```

can NOT be rewritten as

A.

```
while ( --counter >= 1 )
    if ( counter % 2 )
        printf("A");
    else
        printf("B");
```

B.

```
while ( counter >= 1 )
    if (counter % 2)
        printf("A");
    else
        printf("B");
    --counter;
```

C.

```
while ( counter > 1 )
{
    --counter;
    if ( counter % 2 )
        printf("A");
    else
        printf("B");
}
```

D.

```
do
{
    --counter;
    printf( counter % 2 ? "A" : "B" );
} while ( counter >= 2 );
```

Your Answer: B;D | Correct Answer: B;D | Correct | Time taken: 02min 43sec | Discuss

Q #11 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C | Medium (33%)

Consider the following program fragment. Which of the following if condition(s) prints GO Classes?

```
int b = 1, c = 1, d = 0;
if (0 && 0 == 0) //first if
    printf("GO Classes");
if (b || --b == 0) //Second if
    printf("GO Classes");
if (c || c-- == 0) //Third if
    printf("GO Classes");
if (d || ++d == 0) //Fourth if
    printf("GO Classes");
```

A. First if condition
B. Second if condition
C. Third if condition
D. Fourth if condition

Your Answer: B;C | Correct Answer: B;C | Correct | Time taken: 01min 23sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Medium (51%)

Let A and B be two unknown 8-bit 2's complement numbers.
We know the results of $A \,\hat{}\, B$ and $A \,\&\, B$ as shown below:

$A \,\hat{}\, B$	00110100
$A \,\&\, B$	11001001

What is the sum $A + B$ expressed in the 8-bit two's complement notation?

A. 11000110

B. 11000010

C. 01000010

D. 11010110

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 01min 34sec | Discuss

Q #13 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C | Hard (11%)

Which of the following if statement(s) below is/are equivalent to the given switch statement (that is, produces the same output under the same conditions)? Assume the **answer** is a previously declared int.

```
switch (answer)
{
    case 0: printf("0 entered"); break;
    case 1: printf("1 entered"); break;
    case 3: printf("3 entered"); break;
    case 5: printf("5 entered"); break;
    default: printf("Other value entered");
}
```

A.

```
if (answer == 0 || 1 || 3 || 5)
    printf("%d entered", answer);
else
    printf("Other value entered");
```

B.

```
if ((answer >= 1 && answer <= 5 &&
        answer % 2 == 1) || (answer == 0))
    printf("%d entered", answer);
else
    printf("Other value entered");
```

C.

```
if (answer == 0)
    printf("0 entered");
if (answer == 1)
    printf("1 entered");
if (answer == 3)
    printf("3 entered");
if (answer == 5)
    printf("5 entered");
if (answer != 0 && answer != 1
        && answer != 3 && answer != 5)
    printf("Other value entered");
```

D.

```
if (answer == 0)
    printf("0 entered");
else if (answer == 1)
    printf("1 entered");
else if (answer == 3)
    printf("3 entered");
else if (answer == 5)
    printf("5 entered");
else if (answer != 0 & 1 & 3 & 5)
    printf("Other value entered");
```

Your Answer: B;C | Correct Answer: B;C;D | Incorrect | Time taken: 03min 40sec | Discuss

Q #14 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C | Hard (28%)

Consider two given C programs. Which of the following is/are true? It is also known that in C programming, && has higher precedence than $||$.

```
Program 1
#include<stdio.h>
void main()
{
    int i = 0;
    if (++i && (i==0))
        printf(“GO Classes\n”);
    else
        printf(“GATEOverflow\n”);
}
```

```
Program 2
#include<stdio.h>
void main()
{
    int x, y, z;
    x = 1;
    y = z = 0;
    ++x || ++y && ++z;
    if (y || z)
        printf(“GO Classes\n”);
    else
        printf(“GATEOverflow\n”);
}
```

A. Program 1 prints GO Classes
B. Program 1 prints GATEOverflow
C. Program 2 prints GO Classes
D. Program 2 prints GATEOverflow

Your Answer: B;D | Correct Answer: B;D | Correct | Time taken: 03min 49sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Medium (37%)

What will be the output of the following C program?

```
#include<stdio.h>
int main(){
int x=3;
int y;
switch(x++){
    x++;
    case 3:
        printf("Three %d",x);
        break;
    case 4:
        printf("Four %d",x);
        break;
    case 5:
        printf("Five %d",x);
        break;
    default: printf("Default %d",x);
    }
}
```

A. Three 3
B. Four 4
C. Three 4
D. Four 5

Your Answer: C | Correct Answer: C | Correct | Time taken: 01min 46sec | Discuss

You're doing good, you can target above 70 percentage!

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